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Nambur, Pedakakani (M), Guntur (Dt) - 522508
Department of CSE - Artificial Intelligence & Machine Learning

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented ☐ Application oriented ☐ Product oriented ☐ Other (please specify) Specify if others:	Batch ID:	A7

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Title of the Project : Hybrid RoBERTa-BiLSTM for Abusive Language Detection

Abstract

The rapid growth of social media platforms such as Twitter (X), Facebook, Instagram, Reddit, and YouTube has led to a significant increase in abusive, offensive, and hateful content, creating serious challenges for online safety and digital communication. Manual identification of abusive content is time-consuming, inconsistent, and impractical due to the enormous volume of user-generated posts. Therefore, there is a growing need for automated systems that can accurately detect abusive language in real time. The primary stakeholders of this project include social media platforms, content moderators, educational institutions, government organizations, and online communities that aim to provide a safer digital environment.

Several machine learning and deep learning techniques, including Support Vector Machines (SVM), Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), Bidirectional Long Short-Term Memory (Bi-LSTM), and attention-based models, have been proposed for abusive language detection. More recently, transformer-based language models have shown superior performance in understanding contextual semantics. However, many existing approaches still struggle to effectively capture both contextual meaning and sequential dependencies, leading to reduced performance on context-dependent or ambiguous abusive expressions. The research paper selected as the foundation for this project demonstrates the effectiveness of Character CNN, Bi-LSTM, and Attention mechanisms but relies on static word embeddings, which limits contextual understanding.

To overcome these limitations, this project proposes a " Hybrid RoBERTa-BiLSTM for Abusive Language Detection " framework for abusive language detection. The proposed approach utilizes the contextual language understanding capability of the RoBERTa transformer to generate dynamic contextual embeddings. These embeddings are processed by a Bidirectional Long Short-Term Memory (Bi-LSTM) network to capture sequential dependencies, followed by a Multi-Head Attention mechanism that emphasizes the most relevant features contributing to abusive language classification. The model will be trained and evaluated using publicly available benchmark datasets and compared with existing approaches using Accuracy, Precision, Recall, and F1-score.

The scope of this project is limited to text-based abusive language detection in English using benchmark datasets. Multimedia content, multilingual detection, and real-time deployment are beyond the scope of the current work. The success of the proposed system will be measured by its ability to improve classification performance over the baseline model while providing a robust and scalable solution for social media content moderation and future research.

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References (Category specific)

1.Nelatoori, K. B., & Kommanti, H. B. (2023). Attention-based bi-lstm network for abusive language detection. IETE Journal of Research, 69(11), 7884-7892.

Project Team Members

S. No.	Regd. No.	Name of the Student	Signature
1	23BQ1A4242	D.Pavani	
2.	23BQ1A4210	A.Arun Pal	
3.	23BQ1A4203	A.Umesh Karthik	
4.	23BQ1A4211	A.Sai Dhanush	

Project Guide

Name: Dr.P.Sri Hari Signature: